

ICT703 – Module 9 Workshop

Learning Objectives

In this Workshop you will:

1. *Practice defining classes and objects*

A class represents an object in the real world – most modern computing works in an Object-Oriented manner, using classes and objects (aka instances of classes) to represent the world around us.

Activity 1: Students in ICT112

Students in ICT112 (First year programming course) have 3 tasks that make up their final grade. Task 1 (worth 30%) is made up of 8 weekly assignments (out of 5 marks). Only the highest 6 count toward the final Task 1 grade. Task 2 is a report worth 40% and Task 3 is a 30% exam.

Students have a first and last name, student ID, Task 1 list, Task 1 score, Task 2 score and Task 3 score.

1. Create a student class with an initialisation method.
2. Create 5 students and 'enroll' them in your ICT112 course. These will be used to test the method below.
3. Create a function `get_task_1_total()`, which will work through a list of up to 8 entries and determine the total of the HIGHEST 6. Your method should check that no individual mark is higher than 5 or lower than 0 & that there are AT MOST 8 marks, if any of these fail – write out an error message identifying which student's marks are incorrect.
4. Create a data structure that could cope with 150 students. Enroll the 5 students from part 2 into this structure.
5. Calculate and print out the final marks for each student and the grade (HD, D, etc.)